Assignment Summary

Which customer features signal credit default? | Give Me Some Credit data

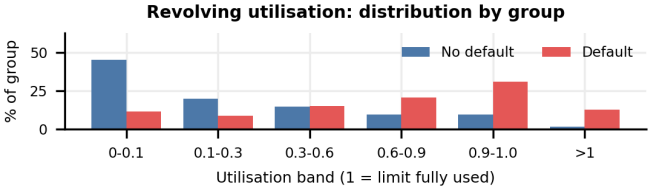
Amirhossein Rezazadeh | 104,797 training customers, 6.6% defaulted | exploratory, no model trained

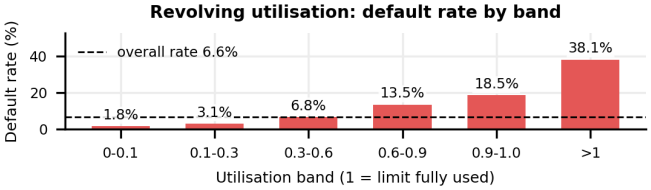
Feature	Shape	% zeros	% missing	% outliers	Median: no default	Median: default
Revolving utilisation	extreme right skew	7.3	0.0	0.5	0.13	0.83
Age	near symmetric	0.0	0.0	0.0	52	46
Monthly income*	extreme right skew	1.4	19.8	4.0	5,500	4,500
90+ days late	zero-inflated count	94.6	0.0	5.4	0 (mean 0.05)	0 (mean 0.67)

**Observed incomes only (19.8% missing). Outliers by the 1.5 x IQR rule. Skewed features are summarised with medians, not means.*

1. Revolving utilisation: defaulters sit closer to their limit

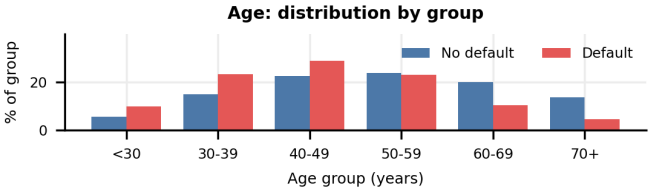
Half of the defaulters use more than 83% of their limit, against 13% for the typical non-defaulter. The default rate rises from 1.8% below 0.1 to 38.1% above 1 (values up to 22,198 are not real ratios). *Why:* a full limit leaves no cushion for a surprise expense.

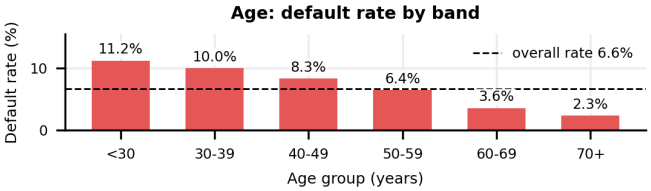




2. Age: younger customers default more often

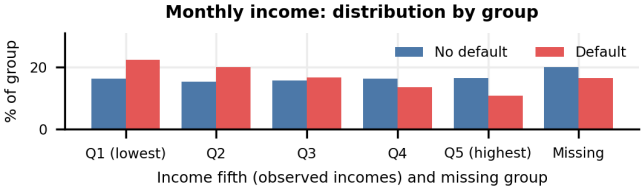
The typical defaulter is 46 and the typical non-defaulter 52; the default rate drops from 11.2% under 30 to 2.3% at 70+. The groups overlap, so age supports other signals. *Why:* shorter credit history and less settled finances.

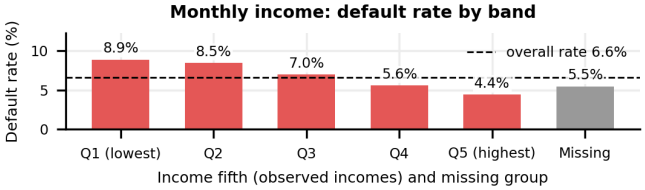




3. Monthly income: skewed, and one in five is missing

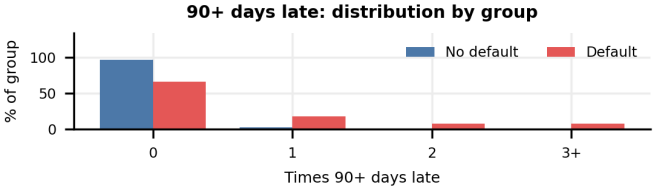
Median income is 4,500 for defaulters and 5,500 for the rest; the default rate falls from 8.9% to 4.4% across income fifths. Customers without income default less (5.5% vs 6.9%) and their DebtRatio is not a real ratio. *Why:* low income leaves less room for shocks, but the signal is moderate.

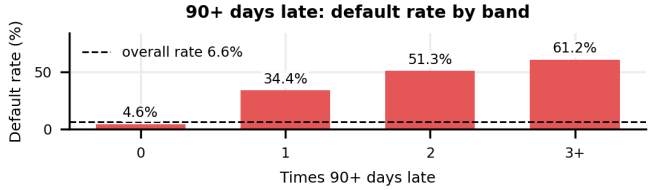




4. Times 90+ days late: rare, but the clearest warning

94.6% of customers were never this late, yet 34.3% of defaulters were at least once (3.4% of non-defaulters). The default rate is 4.6% with no delay and 61.2% with three or more; still, 65.7% of defaulters have none. *Why:* a long delay shows a borrower who cannot, or will not, pay.





Final insight: three features to start with

I would start with **revolving utilisation**, **past-due history** and **age**. Utilisation (AUC 0.78) shows how stretched a borrower is today; past-due history (total late payments, AUC 0.78, led by 90+ day delays) shows how they have actually repaid; age (AUC 0.63) is complete, easy to explain and only weakly related to the others. Income and DebtRatio were left out because they are weaker (AUC 0.57 and 0.53) and have missing or unreliable values.

Default rate = share of a band that defaulted (dashed line: overall 6.6%). AUC = how well one feature alone separates defaulters (0.5 = no information). Training-set figures; associations, not causes. Details: give_me_some_credit_assignment.ipynb